

PL/SQL Exercise 4

Scenario 1: Calculate the age of customers for eligibility checks.

Question: Write a function CalculateAge that takes a customer's date of birth as input and returns their age in years.

```
create or replace function CalculateAge(  
    dob date  
)  
return number  
is  
begin  
    return trunc((sysdate - dob) / 365);  
end;  
/
```

Scenario 2: The bank needs to compute the monthly installment for a loan. Question: Write a function CalculateMonthlyInstallment that takes the loan amount, interest rate, and loan duration in years as input and returns the monthly installment amount.

```
create or replace function CalculateMonthlyInstallment(  
    loan_amount number,  
    interest_rate number,  
    years number  
)  
return number  
is  
    monthly_rate number;  
    months number;  
    emi number;  
begin  
    monthly_rate := interest_rate / (12 * 100);  
    months := years * 12;  
  
    emi := (loan_amount * monthly_rate * power(1 + monthly_rate, months)) /  
        (power(1 + monthly_rate, months) - 1);  
  
    return emi;  
end;  
/
```

Scenario 3: Check if a customer has sufficient balance before making a transaction. Question: Write a function HasSufficientBalance that takes an account ID and an amount as input and returns a boolean indicating whether the account has at least the specified amount.

```
create or replace function HasSufficientBalance(  
    acctId number,  
    amt number  
)  
return boolean  
is  
    bal number;  
begin  
    select balance into bal from accounts  
    where accountid = acctId;  
  
    if bal >= amt then  
        return true;  
    else  
        return false;  
    end if;  
end;
```

```
exception  
    when no_data_found then  
        return false;  
end;  
/
```